

Customer Segmentation & Analysis - SQL, Tableau

Part 1

USE apex;

DROP TABLE IF EXISTS apex_dw;

CREATE TABLE apex_dw AS

SELECT

-- keys

m.member_id,

m.membership_id,

-- Membership attributes --

ms.membership_type,

CASE

WHEN ms.membership_type = 'Individual' THEN 'Single'

WHEN ms.membership_type = 'Family' THEN 'Household'

WHEN ms.membership_type = 'Fleet' THEN 'Corporate'

ELSE 'Premium'

END AS membership_category,

ms.join_date,

YEAR(ms.join_date) AS join_year,

MONTH(ms.join_date) AS join_month,

QUARTER(ms.join_date) AS join_quarter,

-- Geography --

ms.region,

ms.state,

-- Household size --

hh.household_size,

-- Member demographics --

m.gender,

CASE

WHEN m.age < 20 THEN 'Under 20'

WHEN m.age BETWEEN 20 AND 34 THEN '20-34'

WHEN m.age BETWEEN 35 AND 49 THEN '35-49'

WHEN m.age BETWEEN 50 AND 64 THEN '50-64'

ELSE '65+'

END AS age_band,

-- Promo & special flags --

CASE WHEN p1.member_id IS NOT NULL THEN 1 ELSE 0 END AS promo1_flag,

```

CASE WHEN p2.member_id IS NOT NULL THEN 1 ELSE 0 END AS promo2_flag,
CASE WHEN sp.membership_id IS NOT NULL THEN 1 ELSE 0 END AS
special_flag,
-- Service aggregates by member --
-- Maintenance
COALESCE(mai.maintenance_txn_count, 0) AS maintenance_txn_count,
COALESCE(mai.maintenance_spend, 0) AS maintenance_spend,
-- Detailing
COALESCE(det.detailing_txn_count, 0) AS detailing_txn_count,
COALESCE(det.detailing_spend, 0) AS detailing_spend,
-- Tires
COALESCE(ti.tires_txn_count, 0) AS tires_txn_count,
COALESCE(ti.tires_spend, 0) AS tires_spend,
-- Inspections
COALESCE(ins.inspections_txn_count, 0) AS inspections_txn_count,
COALESCE(ins.inspections_spend, 0) AS inspections_spend,
-- Roadside
COALESCE(rs.roadside_txn_count, 0) AS roadside_txn_count,
COALESCE(rs.roadside_spend, 0) AS roadside_spend,
-- Accessories
COALESCE(ac.accessories_txn_count, 0) AS accessories_txn_count,
COALESCE(ac.accessories_spend, 0) AS accessories_spend,
-- Total spend across all service categories --
COALESCE(mai.maintenance_spend, 0)
+ COALESCE(det.detailing_spend, 0)
+ COALESCE(ti.tires_spend, 0)
+ COALESCE(ins.inspections_spend, 0)
+ COALESCE(rs.roadside_spend, 0)
+ COALESCE(ac.accessories_spend, 0) AS total_spend
FROM members m
JOIN memberships ms
ON m.membership_id = ms.membership_id
-- household size per membership --
LEFT JOIN (
    SELECT membership_id,
           COUNT(*) AS household_size
    FROM members
    GROUP BY membership_id
) AS hh
ON m.membership_id = hh.membership_id

```

-- promos & special --

```
LEFT JOIN promoone AS p1 ON m.member_id = p1.member_id
LEFT JOIN promotwo AS p2 ON m.member_id = p2.member_id
LEFT JOIN special AS sp ON ms.membership_id = sp.membership_id
```

-- maintenance aggregate per member --

```
LEFT JOIN (
    SELECT member_id,
           COUNT(*) AS maintenance_txn_count,
           SUM(amount_spent) AS maintenance_spend
    FROM maintenance
    GROUP BY member_id
```

```
) AS mai
ON m.member_id = mai.member_id
```

-- detailing aggregate per member --

```
LEFT JOIN (
    SELECT member_id,
           COUNT(*) AS detailing_txn_count,
           SUM(amount_spent) AS detailing_spend
    FROM detailing
    GROUP BY member_id
```

```
) AS det
ON m.member_id = det.member_id
```

-- tires aggregate per member --

```
LEFT JOIN (
    SELECT member_id,
           COUNT(*) AS tires_txn_count,
           SUM(amount_spent) AS tires_spend
    FROM tires
    GROUP BY member_id
```

```
) AS ti
ON m.member_id = ti.member_id
```

-- inspections aggregate per member --

```
LEFT JOIN (
    SELECT member_id,
           COUNT(*) AS inspections_txn_count,
           SUM(amount_spent) AS inspections_spend
    FROM inspections
    GROUP BY member_id
```

```
) AS ins
ON m.member_id = ins.member_id
```

-- roadside aggregate per member --

```
LEFT JOIN (
  SELECT member_id,
    COUNT(*) AS roadside_txn_count,
    SUM(amount_spent) AS roadside_spend
  FROM roadside
  GROUP BY member_id
) AS rs
ON m.member_id = rs.member_id
```

-- accessories aggregate per member --

```
LEFT JOIN (
  SELECT member_id,
    COUNT(*) AS accessories_txn_count,
    SUM(amount_spent) AS accessories_spend
  FROM accessories
  GROUP BY member_id
) AS ac
ON m.member_id = ac.member_id;
```

member_id	membership_id	membership_type	membership_category	join_date	join_year	join_month	join_quarter	region	state	household_size	gender	age_band	promo1_flag	promo2_flag	special_flag	maintenance_txn_count
MBR00001	MS00001	Individual	Single	2020-12-11	2020	12	4	Southwest	TX	1	F	35-49	0	0	1	3
MBR00002	MS00002	Premium	Premium	2020-03-18	2020	3	1	Midwest	IL	2	F	20-34	1	1	1	0
MBR00003	MS00002	Premium	Premium	2020-03-18	2020	3	1	Midwest	IL	2	M	35-49	1	0	1	2
MBR00004	MS00003	Fleet	Corporate	2022-05-04	2022	5	2	Southeast	FL	2	M	35-49	0	0	1	3
MBR00005	MS00003	Fleet	Corporate	2022-05-04	2022	5	2	Southeast	FL	2	F	35-49	1	1	1	4
MBR00006	MS00004	Family	Household	2022-01-30	2022	1	1	Southwest	AZ	3	F	35-49	0	1	1	2
MBR00007	MS00004	Family	Household	2022-01-30	2022	1	1	Southwest	AZ	3	M	35-49	0	0	1	4
MBR00008	MS00004	Family	Household	2022-01-30	2022	1	1	Southwest	AZ	3	F	35-49	0	1	1	5
MBR00009	MS00005	Individual	Single	2021-11-24	2021	11	4	Southwest	AZ	1	F	35-49	1	1	1	3
MBR00010	MS00006	Individual	Single	2021-03-09	2021	3	1	Northeast	PA	1	F	35-49	0	0	1	0
MBR00011	MS00007	Individual	Single	2020-11-14	2020	11	4	West	CA	1	F	20-34	1	0	1	2
MBR00012	MS00008	Premium	Premium	2024-08-21	2024	8	3	Southwest	TX	2	F	35-49	0	1	1	1
MBR00013	MS00008	Premium	Premium	2024-08-21	2024	8	3	Southwest	TX	2	F	35-49	1	0	1	1
MBR00014	MS00009	Family	Household	2020-09-27	2020	9	3	West	CA	4	M	35-49	1	1	1	4
MBR00015	MS00009	Family	Household	2020-09-27	2020	9	3	West	CA	4	F	Under 20	0	0	1	2
MBR00016	MS00009	Family	Household	2020-09-27	2020	9	3	West	CA	4	M	20-34	0	0	1	1
MBR00017	MS00009	Family	Household	2020-09-27	2020	9	3	West	CA	4	M	35-49	0	1	1	3
MBR00018	MS00010	Fleet	Corporate	2023-08-03	2023	8	3	Southwest	TX	2	F	35-49	0	1	1	4
MBR00019	MS00010	Fleet	Corporate	2023-08-03	2023	8	3	Southwest	TX	2	M	35-49	1	0	1	6
MBR00020	MS00011	Individual	Single	2020-04-08	2020	4	2	Midwest	OH	1	M	20-34	0	0	1	5
MBR00021	MS00012	Premium	Premium	2020-04-02	2020	4	2	Southeast	FL	2	M	50-64	0	1	1	0
MBR00022	MS00012	Premium	Premium	2020-04-02	2020	4	2	Southeast	FL	2	F	35-49	1	1	1	3
MBR00023	MS00013	Fleet	Corporate	2020-10-18	2020	10	4	West	CA	2	F	50-64	0	0	1	4
MBR00024	MS00013	Fleet	Corporate	2020-10-18	2020	10	4	West	CA	2	F	Under 20	0	0	1	2
MBR00025	MS00014	Individual	Single	2021-11-10	2021	11	4	West	CA	1	F	50-64	0	0	1	3

maintenance_spend	detailing_txn_count	detailing_spend	tires_txn_count	tires_spend	inspections_txn_count	inspections_spend	roadside_txn_count	roadside_spend	accessories_txn_count	accessories_spend	total_spend
176.64	0	0.00	1	112.69	0	0.00	0	0.00	1	57.62	346.95
0.00	3	167.31	1	99.96	0	0.00	1	13.72	0	0.00	280.99
77.34	6	309.21	0	0.00	1	17.30	0	0.00	1	83.74	487.59
146.56	0	0.00	2	153.89	4	112.66	2	25.63	0	0.00	438.74
92.20	0	0.00	1	75.75	2	73.68	0	0.00	0	0.00	241.63
110.37	2	61.37	1	62.40	1	49.88	0	0.00	1	31.99	316.01
163.77	1	54.38	0	0.00	1	21.29	0	0.00	0	0.00	239.44
263.21	4	118.42	0	0.00	0	0.00	0	0.00	0	0.00	381.63
116.36	2	91.30	1	60.79	0	0.00	1	29.73	0	0.00	298.18
0.00	0	0.00	1	82.67	3	77.81	1	49.64	0	0.00	210.12
69.55	1	19.06	2	86.25	0	0.00	1	10.75	0	0.00	185.61
29.72	6	265.43	1	115.77	0	0.00	0	0.00	1	16.28	427.20
41.71	5	223.03	0	0.00	1	25.49	0	0.00	1	33.49	323.72
208.72	1	36.42	2	148.74	1	32.27	0	0.00	1	92.04	518.19
117.82	1	18.56	0	0.00	3	87.71	0	0.00	1	31.09	255.18
52.20	0	0.00	2	138.04	0	0.00	0	0.00	1	49.55	239.79
162.10	0	0.00	1	114.19	2	73.58	0	0.00	1	41.22	391.09
207.24	2	75.85	1	79.23	3	85.77	0	0.00	1	10.58	458.67
252.58	1	32.05	1	70.11	3	69.45	0	0.00	2	158.39	582.58
189.17	1	30.46	1	83.05	0	0.00	0	0.00	0	0.00	302.68
0.00	1	26.82	2	186.07	4	142.44	0	0.00	2	80.11	435.44
91.51	0	0.00	0	0.00	1	26.61	0	0.00	1	44.95	163.07
213.10	2	47.74	1	52.95	0	0.00	1	17.95	1	59.26	391.00
75.03	2	53.82	2	115.62	2	37.34	1	14.29	0	0.00	296.10
159.93	0	0.00	0	0.00	1	18.35	0	0.00	0	0.00	178.28

Part 2

1st Question addressed: What distinguishes members who participated in special events from non-participants, and are special event attendees more valuable customers?

SELECT

CASE WHEN special_flag = 1 THEN 'Special Event Participant'

ELSE 'Non-Participant' END AS participation_status,

membership_category,

COUNT(*) AS member_count,

ROUND(AVG(total_spend), 2) AS avg_total_spend,

ROUND(AVG(maintenance_spend + tires_spend), 2) AS

avg_maintenance_tires_spend,

ROUND(AVG(detailing_spend + accessories_spend), 2) AS

avg_detail_accessories_spend,

ROUND(AVG(CASE WHEN (maintenance_txn_count > 0) + (detailing_txn_count > 0)

+

(tires_txn_count > 0) + (inspections_txn_count > 0) +

(roadside_txn_count > 0) + (accessories_txn_count > 0) >= 3 THEN 1 ELSE 0

END) * 100, 1) AS pct_multi_service

FROM apex_dw

GROUP BY participation_status, membership_category

ORDER BY participation_status, membership_category;

participation_status ▲ 1	membership_category ▲ 2	member_count	avg_total_spend	avg_maintenance_tires_spend	avg_detail_accessories_spend	pct_multi_service
Special Event Participant	Corporate	142	468.22	316.40	57.91	97.2
Special Event Participant	Household	359	358.29	209.88	98.99	91.4
Special Event Participant	Premium	166	345.32	162.57	148.02	92.8
Special Event Participant	Single	223	228.28	118.02	72.10	78.0

Insights: Special event participants show high engagement with 78-97% using multiple services across all membership tiers. Corporate participants average \$468.22 in spending with the strongest preference for maintenance and tires (\$316.40), while Premium members favor discretionary services like detailing and accessories (\$148.02), suggesting event attendees are highly engaged customers worth targeting for retention programs.

2nd Question addressed:

How do service preferences and spending patterns vary across different age groups, and which age demographics represent the highest value segments?

```
SELECT
    age_band,
    membership_category,
    COUNT(*) AS member_count,
    ROUND(AVG(total_spend), 2) AS avg_total_spend,
    ROUND(AVG(maintenance_spend), 2) AS avg_maintenance,
    ROUND(AVG(detailing_spend), 2) AS avg_detailing,
    ROUND(AVG(tires_spend), 2) AS avg_tires,
    ROUND(AVG(roadside_spend), 2) AS avg_roadside,
    ROUND(AVG(accessories_spend), 2) AS avg_accessories,
    ROUND(AVG(CASE WHEN total_spend > 0 THEN 1 ELSE 0 END) * 100, 1) AS
pct_active
FROM apex_dw
WHERE age_band != 'Unknown'
GROUP BY age_band, membership_category
ORDER BY age_band, membership_category, avg_total_spend;
```

age_band	membership_category	member_count	avg_total_spend	avg_maintenance	avg_detailing	avg_tires	avg_roadside	avg_accessories	pct_active
20-34	Corporate	30	495.37	188.38	33.54	158.42	21.90	12.12	100.0
20-34	Household	121	349.65	103.21	61.49	102.04	14.03	36.66	99.2
20-34	Premium	25	337.58	110.75	84.93	54.65	6.16	53.14	100.0
20-34	Single	84	252.82	90.71	50.80	39.67	7.33	26.27	100.0
35-49	Corporate	70	450.72	170.91	37.50	118.26	21.23	21.25	100.0
35-49	Household	168	359.24	114.63	59.43	98.09	16.08	36.48	100.0
35-49	Premium	84	354.22	104.62	89.14	57.70	3.82	68.52	100.0
35-49	Single	94	211.09	67.80	38.91	43.04	8.72	25.40	98.9
50-64	Corporate	38	479.71	207.76	36.21	132.14	14.51	29.69	100.0
50-64	Household	47	378.86	118.00	63.16	101.33	19.90	41.19	100.0
50-64	Premium	48	339.13	102.81	75.42	55.99	7.62	64.88	100.0
50-64	Single	24	213.36	58.67	39.65	54.10	3.70	27.24	100.0
65+	Household	2	377.49	49.43	142.22	80.34	30.11	22.38	100.0
65+	Premium	8	312.70	142.09	67.81	43.60	0.00	45.01	100.0
65+	Single	1	205.95	75.86	24.05	61.78	17.92	0.00	100.0
Under 20	Corporate	4	461.84	282.37	33.83	59.60	33.31	25.50	100.0
Under 20	Household	21	352.59	92.42	74.54	107.82	14.64	35.68	100.0
Under 20	Premium	1	348.55	41.18	73.52	68.09	0.00	165.76	100.0
Under 20	Single	20	225.02	73.82	46.50	31.33	7.45	49.97	100.0

Insights: Corporate members aged 50-64 show the highest spending at \$479.71, driven primarily by maintenance (\$207.76). Across all age groups, Corporate members consistently spend 2x more than Single members, while the 35-49 age band has the largest member base (416 members), representing the best opportunity for tier upgrade campaigns.

Part 3

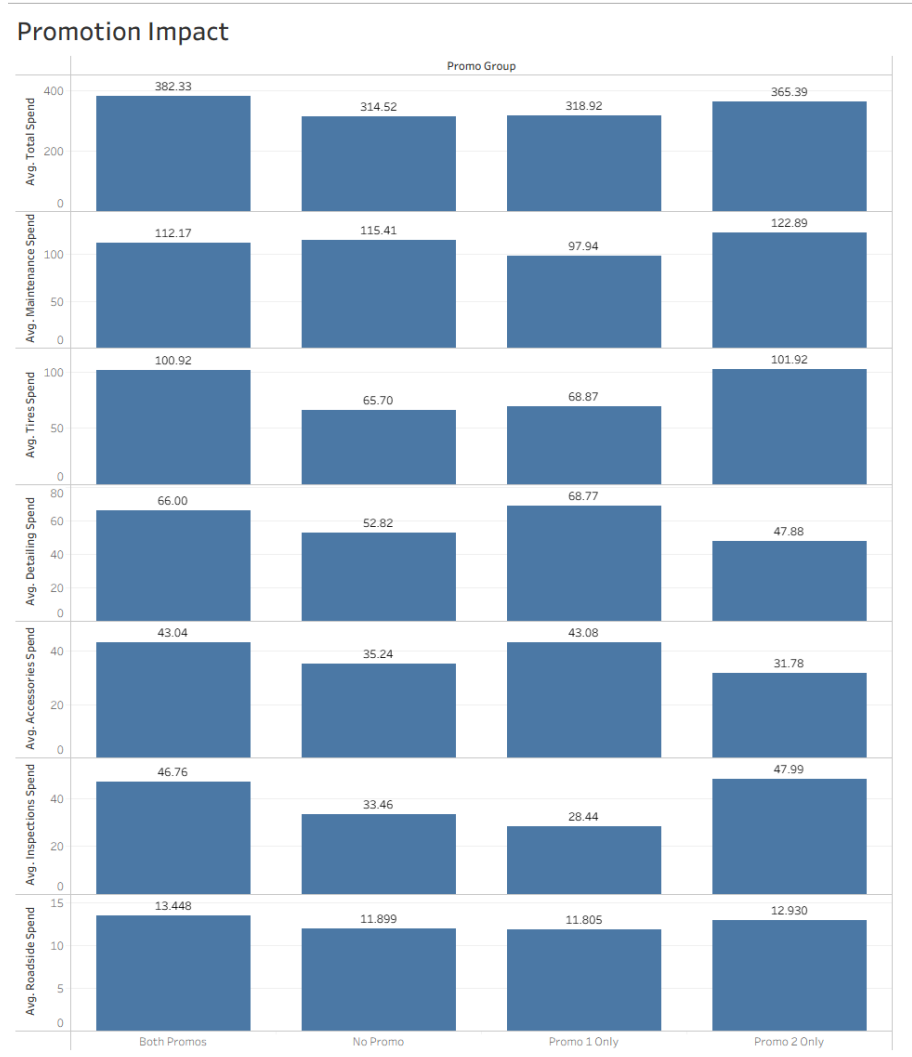
Services spent by Region Visualization



Insights: Regions like the Northeast and Midwest show the highest service spending, while the West lags behind across all categories. This suggests stronger engagement in high-spend regions and untapped potential in lower-spend areas.

Recommendation: Increase marketing and service availability in the West, and reinforce high-margin services (like Detailing) in moderate-spend regions to balance demand.

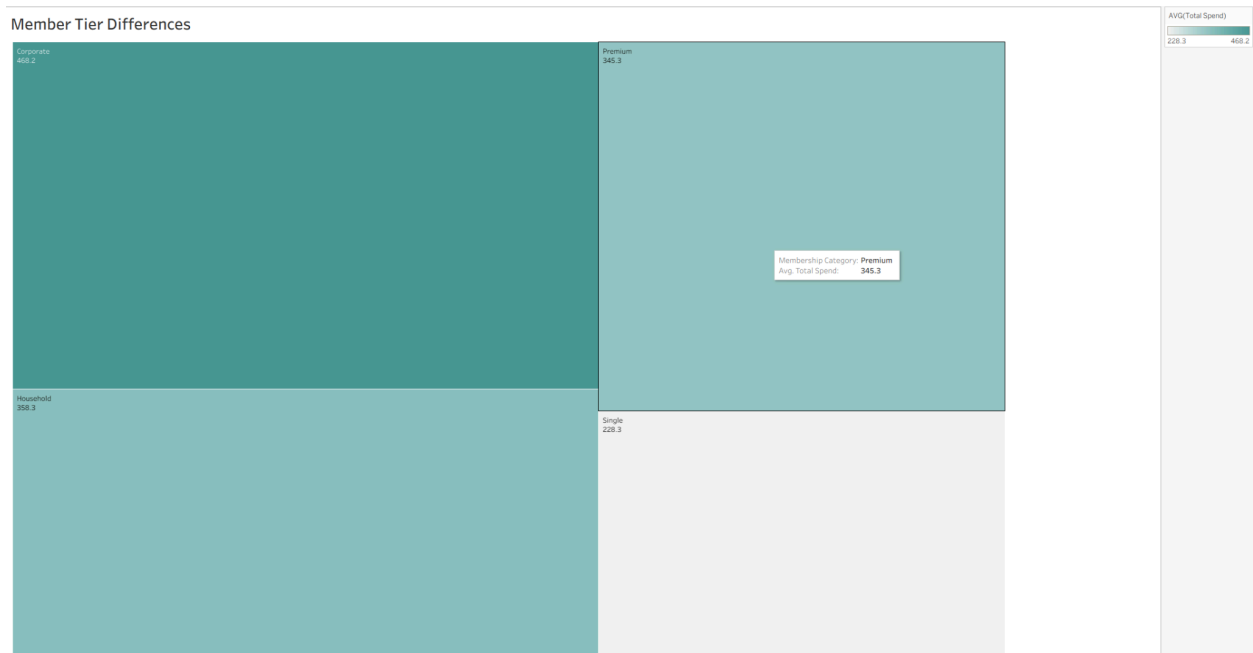
Promotion Impact Visualization



Insights: Members receiving both promotions show the highest total and category-level spend, while non-promo members spend the least. Promotions successfully drive higher engagement.

Recommendation: Continue combined promotions, expand the more effective promo type, and target non-promo members with introductory offers to lift participation.

Member Tier Differences Visualization



Insights: Corporate members spend the most on average, followed by Household and Premium tiers, while Single members contribute the least. Member tier clearly influences overall service value.

Recommendation: Prioritize retention and tailored perks for Corporate members, while offering upgrade incentives and bundled packages to lift spend among Single and Household members.

Part 4

Which service combinations show the strongest cross-selling potential?

```
SELECT
  CASE
    WHEN maintenance_spend > 0 THEN 'Maintenance'
    ELSE 'No Maintenance'
  END AS maintenance_buyer,
  ROUND(AVG(CASE WHEN detailing_spend > 0 THEN 1 ELSE 0 END) * 100, 1) AS
pct_buy_detailing,
  ROUND(AVG(CASE WHEN tires_spend > 0 THEN 1 ELSE 0 END) * 100, 1) AS
pct_buy_tires,
  ROUND(AVG(CASE WHEN inspections_spend > 0 THEN 1 ELSE 0 END) * 100, 1)
AS pct_buy_inspections,
  ROUND(AVG(CASE WHEN roadside_spend > 0 THEN 1 ELSE 0 END) * 100, 1) AS
pct_buy_roadside,
  ROUND(AVG(CASE WHEN accessories_spend > 0 THEN 1 ELSE 0 END) * 100, 1)
AS pct_buy_accessories,
```

```

COUNT(*) AS member_count,
ROUND(AVG(total_spend), 2) AS avg_total_spend
FROM apex_dw
GROUP BY maintenance_buyer
ORDER BY maintenance_buyer;

```

maintenance_buyer	pct_buy_detailing	pct_buy_tires	pct_buy_inspections	pct_buy_roadside	pct_buy_accessories	member_count	avg_total_spend
Maintenance	77.5	61.3	69.7	38.9	54.8	799	356.58
No Maintenance	79.1	51.6	57.1	34.1	61.5	91	202.55

Insights: Maintenance buyers are significantly more engaged, with 77.5% also purchasing detailing and 69.7% purchasing inspections, compared to lower rates for non-maintenance buyers. They also spend 76% more on average (\$356.58 vs \$202.55)

Recommendation: Making Maintenance buyers prime targets for cross-selling and bundled service promotions.

Do larger households use more varied services?

```

SELECT
CASE
    WHEN household_size = 1 THEN '1 member'
    WHEN household_size = 2 THEN '2 members'
    WHEN household_size BETWEEN 3 AND 4 THEN '3-4 members'
    WHEN household_size >= 5 THEN '5+ members'
END AS household_category,
COUNT(*) AS total_members,
ROUND(AVG(total_spend), 2) AS avg_total_spend,
ROUND(AVG(maintenance_txn_count + detailing_txn_count + tires_txn_count +
    inspections_txn_count + roadside_txn_count + accessories_txn_count), 1) AS
avg_transactions,
ROUND(AVG(CASE WHEN maintenance_txn_count > 0 THEN 1 ELSE 0 END) *
100, 1) AS pct_use_maintenance,
ROUND(AVG(CASE WHEN detailing_txn_count > 0 THEN 1 ELSE 0 END) * 100, 1)
AS pct_use_detailing,
ROUND(AVG(CASE WHEN accessories_txn_count > 0 THEN 1 ELSE 0 END) * 100,
1) AS pct_use_accessories,
ROUND(AVG(accessories_spend), 2) AS avg_accessories_spend
FROM apex_dw
GROUP BY
CASE
    WHEN household_size = 1 THEN '1 member'
    WHEN household_size = 2 THEN '2 members'

```

```

        WHEN household_size BETWEEN 3 AND 4 THEN '3-4 members'
        WHEN household_size >= 5 THEN '5+ members'
    END
ORDER BY
    MIN(household_size);

```

household_category	total_members	avg_total_spend	avg_transactions	pct_use_maintenance	pct_use_detailing	pct_use_accessories	avg_accessories_spend
1 member	223	228.28	5.4	84.8	71.3	47.5	28.02
2 members	400	396.80	8.8	93.5	79.3	59.5	43.54
3-4 members	267	351.00	7.7	88.4	80.5	56.2	36.27

Insights: Larger households spend significantly more, with 2-member households averaging \$396.80 compared to single-member households at \$228.28. Two-member households show the highest engagement across all service categories (93.5% use maintenance, 79.3% use detailing) and average 8.8 transactions, making them the most valuable segment despite 3-4 member households having higher accessories spending (\$43.54 vs \$36.27).

Recommendation:

Target 2-member households (likely couples) with bundled maintenance and detailing packages, as they demonstrate the strongest overall engagement and spending. Additionally, promote accessories to 3-4 member families (likely parents with children) who show natural interest in this category but lower overall service utilization.

Part 5

```

SELECT
    'Promo 1' AS promotion,
    CASE WHEN promo1_flag = 1 THEN 'Promo1 Participant' ELSE 'Non-Participant'
END AS participant_status,
    COUNT(*) AS member_count,
    ROUND(AVG(total_spend), 2) AS avg_total_spend,
    ROUND(AVG(maintenance_spend), 2) AS avg_maintenance,
    ROUND(AVG(detailing_spend), 2) AS avg_detailing,
    ROUND(AVG(tires_spend), 2) AS avg_tires,
    ROUND(AVG(inspections_spend), 2) AS avg_inspections,
    ROUND(AVG(roadside_spend), 2) AS avg_roadside,
    ROUND(AVG(accessories_spend), 2) AS avg_accessories,
    ROUND(AVG(CASE WHEN maintenance_txn_count > 0 THEN 1 ELSE 0 END) *
100, 1) AS pct_use_maintenance,
    ROUND(AVG(CASE WHEN detailing_txn_count > 0 THEN 1 ELSE 0 END) * 100, 1)
AS pct_use_detailing,
    ROUND(AVG(CASE WHEN tires_txn_count > 0 THEN 1 ELSE 0 END) * 100, 1) AS
pct_use_tires
FROM apex_dw

```

GROUP BY participant_status

UNION ALL

SELECT

```
'Promo 2' AS promotion,
CASE WHEN promo2_flag = 1 THEN 'Promo2 Participant' ELSE 'Non-Participant'
END AS participant_status,
COUNT(*) AS member_count,
ROUND(AVG(total_spend), 2) AS avg_total_spend,
ROUND(AVG(maintenance_spend), 2) AS avg_maintenance,
ROUND(AVG(detailing_spend), 2) AS avg_detailing,
ROUND(AVG(tires_spend), 2) AS avg_tires,
ROUND(AVG(inspections_spend), 2) AS avg_inspections,
ROUND(AVG(roadside_spend), 2) AS avg_roadside,
ROUND(AVG(accessories_spend), 2) AS avg_accessories,
ROUND(AVG(CASE WHEN maintenance_txn_count > 0 THEN 1 ELSE 0 END) *
100, 1) AS pct_use_maintenance,
ROUND(AVG(CASE WHEN detailing_txn_count > 0 THEN 1 ELSE 0 END) * 100, 1)
AS pct_use_detailing,
ROUND(AVG(CASE WHEN tires_txn_count > 0 THEN 1 ELSE 0 END) * 100, 1) AS
pct_use_tires
FROM apex_dw
GROUP BY participant_status
ORDER BY promotion, participant_status DESC;
```

participant_status	2	member_count	avg_total_spend	avg_maintenance	avg_detailing	avg_tires	avg_inspections	avg_roadside	avg_accessories	pct_use_maintenance	pct_use_detailing	pct_use_tires
Promo1 Participant		352	348.82	104.65	67.46	83.98	37.08	12.58	43.06	88.4	80.1	59.1
Non-Participant		538	335.61	118.51	50.77	80.71	39.48	12.33	33.81	90.7	76.0	61.2
Promo2 Participant		389	372.62	118.32	55.61	101.49	47.46	13.15	36.59	90.2	75.8	60.9
Non-Participant		501	316.16	108.92	58.74	66.87	31.60	11.86	38.15	89.4	79.0	59.9

Insights are drawn from both the above query and the visualization in Question 3

Promo1 primarily targeted detailing services, with participants spending significantly more on detailing (\$67.46 vs \$50.77) and showing higher usage rates (80.1% vs 76.0%). Interestingly, Promo1 participants spent less on maintenance (\$104.65 vs \$118.51), suggesting the promotion successfully shifted focus toward detailing services.

Promo2 strongly influenced tires and accessories, with participants spending notably more on tires (\$101.49 vs \$66.87) and accessories (\$36.59 vs \$38.15), while also increasing tire purchase rates (60.9% vs 59.9%). Total spending is higher for Promo2 participants (\$372.62 vs \$316.16), indicating this was a more revenue-effective promotion overall.